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BJSM Online First, published on July 27, 2016 as 10.1136/bjsports-2016-096040

How do training and competition workloads relate to injury? The workload—injury aetiology model

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Accepted 30 May 2016

ABSTRACT

Injury aetiology models that have evolved over the previous two decades highlight a number of factors which contribute to the causal mechanisms for athletic injuries. These models highlight the pathway to injury, including (1) internal risk factors (eg, age, neuromuscular control) which predispose athletes to injury, (2) exposure to external risk factors (eg, playing surface, equipment), and finally (3) an inciting event, wherein biomechanical breakdown and injury occurs. The most recent aetiological model proposed in 2007 was the first to detail the dynamic nature of injury risk, whereby participation may or may not result in injury, and participation itself alters injury risk through adaptation. However, although training and competition workloads are strongly associated with injury, existing aetiology models neither include them nor provide an explanation for how workloads alter injury risk. Therefore, we propose an updated injury aetiology model which includes the effects of workloads. Within this model, internal risk factors are differentiated into modifiable and non-modifiable factors, and workloads contribute to injury in three ways: (1) exposure to external risk factors and potential inciting events, (2) fatigue, or negative physiological effects, and (3) fitness, or positive physiological adaptations. Exposure is determined solely by total load, while positive and negative adaptations are controlled both by total workloads, as well as changes in load (eg, the acute:chronic workload ratio). Finally, we describe how this model explains the load—injury relationships for total workloads, acute: chronic workload ratios and the training load—injury paradox.

INTRODUCTION

Athletic injuries are common in a variety of sports,[1 2] compromising performance,[3–5] posing a financial burden to organisations,[6] and potentially causing long-term health consequences.[7] The causes for these injuries are numerous, highlighted by a number of multifactorial injury aetiology models.[8–10] However, regardless of the interplay of risk factors or inciting biomechanical event, every athletic injury is sustained while athletes are exposed to training and competition workloads. Match workloads are due to the competitive demands of the sport, while practice workloads are applied to athletes with the goal of inducing positive physiological changes and maximising performance.

While adequate workloads are necessary for fitness and performance improvements through adaptation and skill acquisition, high workloads and especially 'spikes' in loads are strongly associated with injuries.[3 11–14] Although this association has been demonstrated in soccer,[15] Australian Football,[16–18] basketball,[19] rugby union,[3] rugby league[13 20 21] and cricket players,[12 22 23] few people have explained the underlying mechanisms through which workloads relate to injuries. Moreover, existing injury aetiology models do not account for the relationship between workloads and injury risk. Therefore, the purpose of this paper is to present an updated aetiology model that accounts for the effects of workloads on athletic injuries. After reviewing previous injury aetiology models, we discuss the known association of workloads and injuries, and present an updated model that explains a threefold contribution of workloads to injury risk. Finally, we discuss the research and practice implications of the updated model.

WHY DO ATHLETES GET HURT? INJURY AETIOLOGY MODELS

The multifactorial nature of athletic injuries is well known.[8 10 24] Single risk factors fail to adequately predict injuries, and the limitations of univariate analyses of risk factors have been discussed.[8] Instead, multivariate techniques are recommended, which enable researchers to examine multiple risk factors which come together to cause an injury.[8 24] The first of these multifactorial models was proposed by Meeuwisse in 1994.[8] Meeuwisse built on chronic disease models to propose an epidemiological, multifactorial approach to understanding injury causation and investigating athletic injuries (figure 1). Within this multifactorial model, athletes have intrinsic (internal) risk factors which inherently predispose them to injury. Some of these factors (eg, age) are non-modifiable, while others (eg, flexibility) are modifiable. These predisposed athletes are then exposed to extrinsic (external) risk factors, such as the playing surface, protective equipment, or opponent behaviour, which make them susceptible to an injury. Finally, a certain inciting event occurs wherein the biomechanical stress of the event exceeds the tolerance of the athlete's tissues, and injury ensues. Lastly, Meeuwisse also acknowledged that risk factors are interrelated, and in a subsequent article addressed the importance of interpreting interaction and confounding effects in injury aetiology research.[25]

In two subsequent papers published in *BJSM*, Bahr and colleagues[9 24] built on this first multifactorial model, highlighting and focussing on certain aspects (figure 2). First, they expanded the description of internal and external risk factors and the subsequent methodological implications.[24] Second, they proposed a more thorough description of the biomechanical factors contributing to the inciting event,[24] based on the comprehensive biomechanical injury model of McIntosh.[26]

To cite: Windt J, Gabbett TJ. *Br J Sports Med* Published Online First: [*please include* Day Month Year] doi:10.1136/bjsports-2016-096040

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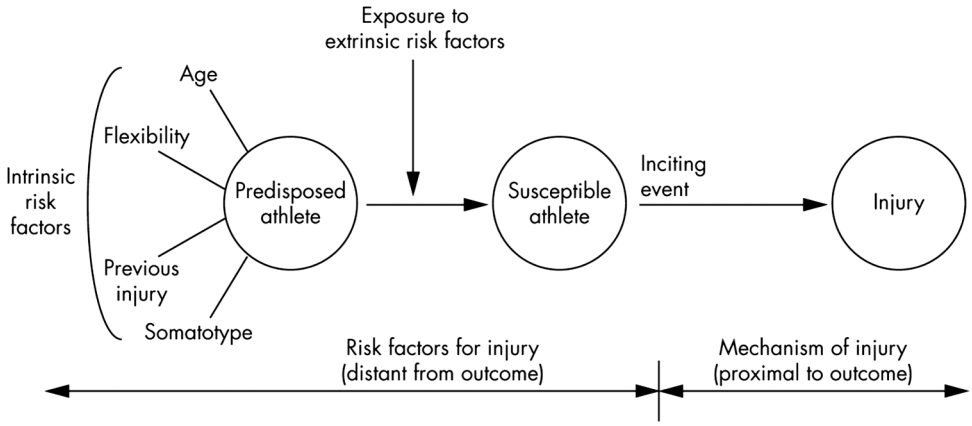

Figure 1 Meeuwisse's first multifactorial injury aetiology model proposed in 1994.[8]

Most recently, Meeuwisse *et al*[10] proposed a modification to these aetiology models in the form of a 'dynamic, recursive injury etiology model' (figure 3). This dynamic model of risk and causation acknowledges that previous linear models do not account for the dynamic and non-linear nature of athletes' injury risk. Moreover, sport participation may or may not result in an injury occurrence. In cases where participation does not result in injury, it actually modifies certain risk factors through physiological adaptation to the training stress. On the other hand, if athletes do experience an injury they may either recover and return to play with a modified injury risk, or never recover, resulting in complete removal from participation.

This continued evolution in understanding injury aetiology helps to inform injury prevention efforts. It provides a framework to inform study design, indicates appropriate statistical analyses, highlights important variables to consider, and in the case of the dynamic model, reminds researchers that participation not only predisposes to injury, but modifies subsequent risk. Further, the consistent modification of these aetiology models highlights the need for further research, and willingness to amend the model as more knowledge is accumulated.[27]

Notably, workloads are not included as an internal risk factor nor an external risk factor in existing aetiology models. In fact, although workloads are strongly associated with injury,[3 11–13] their role in the aetiology of athletic injuries is uncertain, given that they are not explicitly included anywhere within the previous models.

WORKLOADS AND INJURIES—WHAT DO WE KNOW?

Defining and quantifying workloads

Workloads, as defined in Gabbett *et al*'s[15] systematic review on workloads, performance and injury, are 'the cumulative amount of stress placed on an individual from multiple training sessions

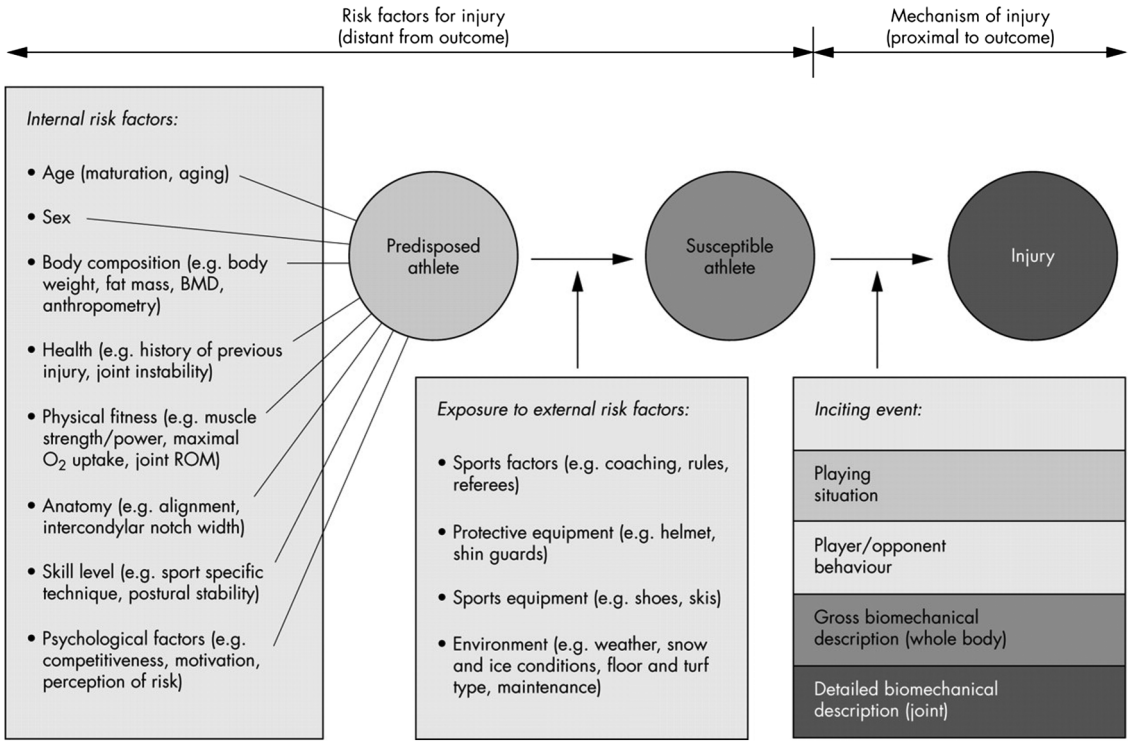

Figure 2 Bahr and Krosshaug's[9] comprehensive aetiology model. BMD, bone mineral density, ROM, range of motion.[9]

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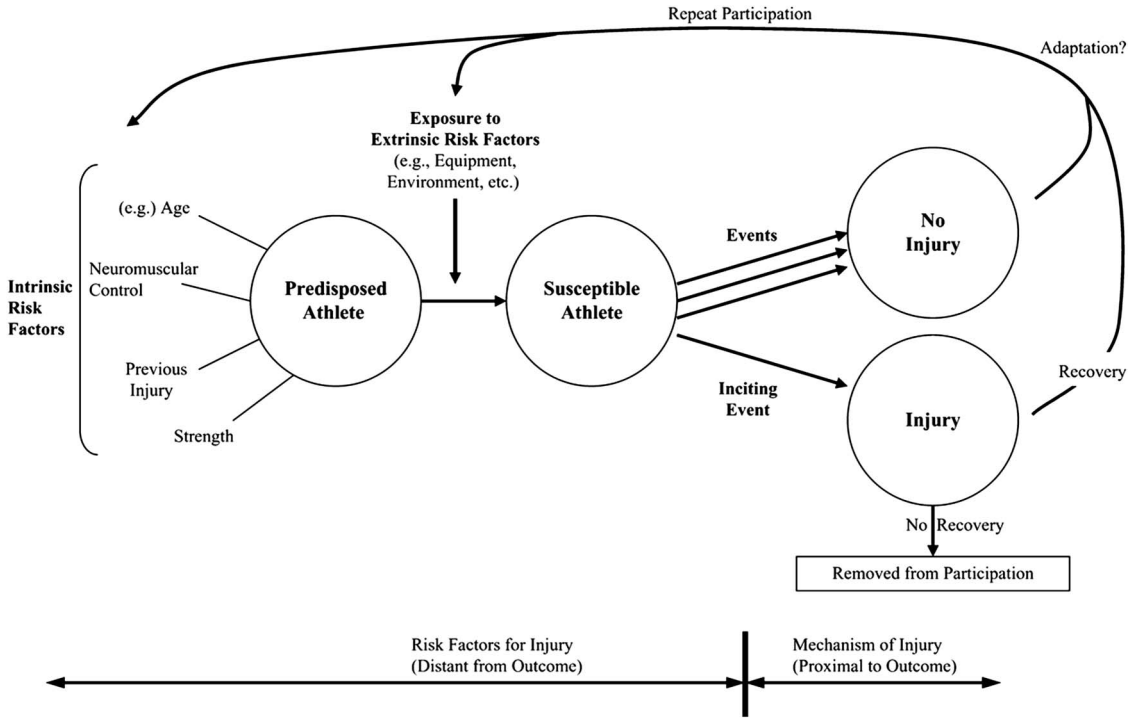

Figure 3 Meeuwisse *et al*'s[10] dynamic, recursive athletic injury aetiology model.

and games over a period of time'. As Smith explains, they are 'a combination of … [training and competition] intensity, duration and frequency'.[28] Essentially, workloads quantify the demands imposed on an athlete during one or more matches or training bouts. Ideally, training is prescribed in such a way that the athletes' homeostasis is disrupted, and optimal adaptation occurs during recovery. This fine balance seeks to avoid, on the one hand—insufficient workloads that fail to induce adaptation or result in detraining, and on the other hand—excessive loads that induce maladaptation or overtraining.[28–30] However, optimally prescribing these loads presents a number of challenges, the most notable of which is choosing an appropriate workload measure.

Workloads can be measured as either external or internal loads. External loads quantify the amount of work performed by the athlete (eg, distance covered, balls thrown, etc), while internal loads measure the 'relative physiological and psychological stress imposed' on the athlete.[29] A given external load will elicit different internal responses in each athlete, based on the characteristics of how the external load is applied and the athlete's individual characteristics (eg, genetics, fitness level, training background, etc).[31] Numerous measures have been developed and proposed for internal and external loads from which practitioners can decide (table 1).[29 32] However, each of these measures aims to quantify the workload completed by the athlete and/or their response to that work. Careful consideration must be given to which type (internal or external) and specific measure(s) of load are most appropriate, given the sport context, goals of load monitoring, logistical and financial constraints, and the psychometric properties (validity/reliability) of the specific measure.

Monitoring workloads in relation to performance, periodisation and overtraining

As Borresen and Lambert[32] describe in their review of workloads, physiological responses and performance, 'optimizing training first involves quantifying what the athlete is currently doing'. Traditionally, this quantification of load has allowed, (1) performance to be predicted, (2) training to be planned, periodised, and tracked, and (3) training readiness and stress to be monitored.

1. *Predicting performance*. In the mid-1970s, Banister *et al*[33] proposed the first systems model of performance, characterising the athlete as a system with input (workload) and subsequent output (performance). Their landmark model proposed fitness as a positive impulse of training, and fatigue as a negative impulse, acknowledging the dual forces of stress/breakdown caused by a training bout and the potential for positive adaptation with sufficient recovery. Within this model, both impulses decline exponentially once a session ends and recovery begins, but fatigue decays at a much more rapid rate than fitness.

These basic principles underpin performance strategies like tapering, where a 3% (ranging 0.5–6.0%) improvement in performance may be attained through altering training volume and intensity to minimise fatigue while maintaining physiological

Table 1 Measures of external and internal load

External load	Internal load
▶ Frequency (/day/week/year)	▶ Session-rating of perceived exertion (RPE*minutes)
▶ Time (minutes, seconds)	▶ HR:RPE ratio
▶ Accelerometer loads	▶ TRIMP (HR-based training impulse)
▶ Distance covered	▶ Blood lactate
▶ High speed distance covered	▶ Lactate:RPE ratio
▶ Jumps completed	▶ Recovery/stress/wellbeing questionnaires (eg, Recovery Stress Questionnaire for Athletes—REST-Q; Daily Analysis of Life Demands for Athletes—DALDA, Profile of Mood States—POMS)
▶ Power output, speed, acceleration	
▶ Time-motion analysis	
▶ Neuromuscular function	
▶ Weight lifted	
▶ Throws/pitches/bowls performed	

Various examples for quantifying both external and internal workload.[29 32 79]

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adaptations, thereby optimising performance at peak competitions.[34–36] While attractive in theory and implemented widely in practice, Banister's model and similar performance prediction models[37 38] have some limitations in their ability to achieve high levels of precision.[39] These limitations may be partly attributable to individually variable load—adaptation responses to set training regimes.[40 41] It is thus important that these basic principles should be adapted to individual athletes, with the applied load and individual response monitored where possible.[42]

2. *Planning, periodising and implementing training plans*. Coaches use the inherent principles of workloads and adaptation to design periodised training plans, with the goal of eliciting optimal adaptation.[43 44] However, workloads must be monitored if coaches are to ensure that their training plan is implemented as intended, as the discrepancy between coaches' and athletes' perceptions of training has been demonstrated.[45] Further, workload monitoring is necessary to make and implement refined alterations to training loads. For example, these precise refinements are implemented during tapering phases where training volume may be exponentially reduced while intensity remains high, with the goal of minimising cumulative fatigue and peaking for competition.[34–36]
3. *Monitoring fatigue to prevent overtraining*. Monitoring workloads over time can also be utilised to assess athletes' stress levels, fatigue, mood and readiness to train.[29] Both physiological measures (eg, heart rate variability, heart rate recovery, hormone levels, catecholamine levels) and psychological measures (eg, REST-Q, POMS, DALDA) have been recommended to monitor athletes' internal workloads and responses.[30 46–48] In monitoring workloads longitudinally, large individual deviations from normal responses, and discrepancies between internal and external load measures can be used to assess athletes' responses to training.[29] For example, lower internal loads with a constant external load may indicate fitness improvements, while increased internal loads with the same external load may indicate a state of fatigue.[29]

In all of these traditional uses, training loads are imposed on athletes with the goal of maximising performance, acknowledging that while training is necessary to induce the 'reward' of positive adaptations, it carries the known 'risk' of imposing stress on the athlete. Thus, every time athletes train, they are exposed to the fatiguing effects of training, as well as risks of potential maladaptation through overtraining. However, excluded in these traditional uses and this aforementioned risk—reward balance is the reality that every training and competition load carries the potential for athletic injury.[49] In fact, every athletic injury is sustained while an athlete performs some sort of workload. This begs the question: how do workloads relate to injury occurrence? Is it a simple matter of increased loads leading to increased injury, or is there more at play? Recently, researchers have begun providing some answers in numerous sports and with various load measures to tease out this association.

Total workloads and injuries

With each training and competition bout, athletes are exposed to the risk of sustaining an injury, so it may seem intuitive that increased loads should result in increased injury incidence. Traditionally, workload-injury investigations focused on this relationship between absolute workloads and injury.[19 20 50–52] To highlight a few examples, rugby players running more than 9 m of very high speed running were 2.7 times more likely to experience a non-contact injury than those below that threshold.[53] Australian football players with higher total distance (OR 5.5) and sprint distance (OR 3.7) accumulated over 3 weeks were at a greater likelihood of injury.[16] Furthermore, higher pitching loads in baseball,[54 55] and bowling loads in cricket have also been associated with increased injury risk.[22]

Gabbett[11] developed an injury prediction model using session-rating of perceived exertion as a marker of internal load among elite rugby league players over 2 years. During the subsequent 2 years, the model was used to predict injuries, showing that players who exceeded the weekly workload threshold as determined by the model were 70 times more likely to test positive for non-contact, soft-tissue injuries, while players who did not exceed the threshold were injured 1/10 as often.[11] Collectively, these data indicate that there is an increase in injury risk with absolute workload increases.

This higher workload—higher injury relationship may create the impression that workloads should be kept low to minimise injury risk. While a reduction in training load may be appropriate in certain instances,[51] two important points should be stated. First, adequate workloads are necessary to induce beneficial physiological adaptations such as high aerobic capacity, optimal body composition, strength, and repeat-sprint ability,[56–58] which are required for high performance, and many of which are associated with decreased injury risk.[53 59 60] Thus, workloads that are too low may not only decrease performance, but may result in lower levels of fitness and preparedness, subsequently increasing injury risk. For example, a 'dual threshold' has been shown in cricket fast bowlers, where not only high workloads and short recovery periods were associated with increased injury risk, but a low total number of deliveries (<123, relative risk=1.4) and long recovery periods (>5 days, relative risk=1.8) also increased injury risk.[22]

Second, it may not solely be the total workload that is applied that contributes to injury risk, but the way in which it is applied. Indeed, some investigations show that total workloads are not always associated with increased injury risk, while the rate of change in these workloads over time is a stronger predictor.[12 13 61]

Acute: chronic workload ratio and injuries

Specifically, load-injury investigations have recently examined the relationship between acute (1-week) and chronic (4-week) workloads, termed the *acute:chronic workload ratio,* with injury risk.[3 12 13] In the first study of acute:chronic workload ratios, Hulin *et al*[12] showed that while internal and external loads of cricket fast bowlers in isolation were not significantly related to injury risk, acute:chronic workload ratios of >1.5 increased injury risk by 2–4 times in the subsequent 7 days.

High acute:chronic ratios, simply described as training 'spikes', similarly increase injury risk in rugby league players[13 62] and professional soccer players,[63] (figure 4) as well as subsequent injury during athletes' return to play[64] (figure 5). In addition to understanding that both total workloads and changes in load over time are related to injury risk, researchers have begun combining these two variables to determine the optimal outcome for injuries and performance.

Workload—injury paradox

In contrast to the idea that higher workloads contribute to higher injury incidence, high workloads may contribute to well-developed physical qualities, thereby reducing injury risk.[61 65 66] The training—injury prevention paradox model has

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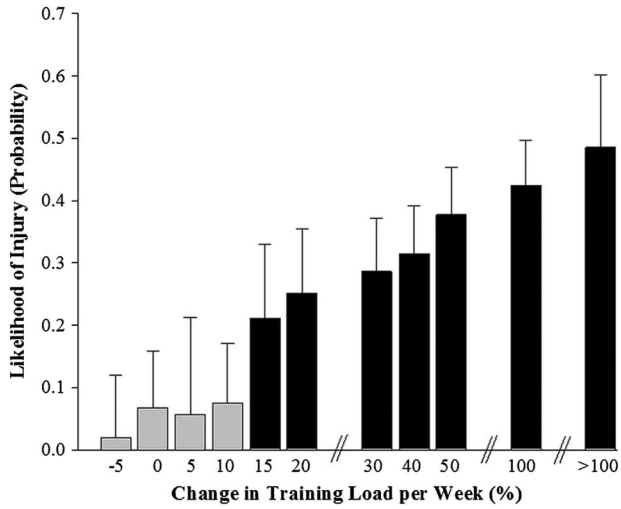

Figure 4 Injury likelihood associated with various weekly changes in workloads. Unpublished data collected from professional rugby league players over three preseason preparation periods. Loads were measured using session-rating of perceived exertion, and a total of 148 injuries were sustained. Reused, with permission, from Gabbett.[61]

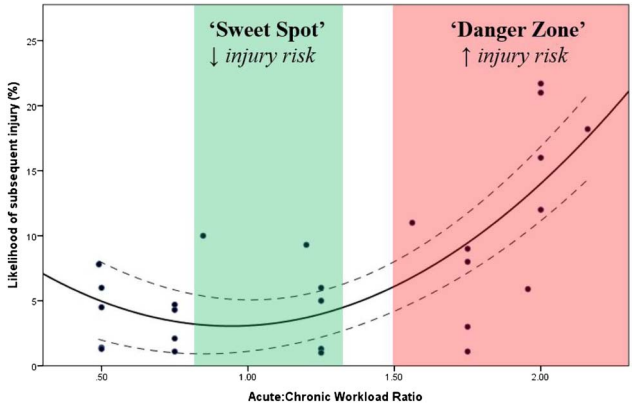

Figure 5 Finding the 'sweet spot' for acute:chronic workload ratios in return-to-play protocols. Data redrawn from 3 separate sports (cricket, rugby league and Australian football) demonstrating injury likelihoods associated with various acute:chronic workload ratios. With permission, from Gabbett.[61]

recently been defined and reviewed by Gabbett[61]—describing this chronic load effect. This 'paradox' of high chronic workloads is their potential preventative effect, as long as week-to-week load changes are kept within ~10% and the acute:chronic workload ratio is kept in a moderate range (ie, training spikes are avoided).

Hulin *et al* found that elite rugby players who had very high acute:chronic workload ratios as well as high chronic workloads had the largest risk of injury (table 2). However, they also demonstrated that as long as acute:chronic workload ratios were kept within a moderate zone (0.85–1.35), high chronic workloads were associated with the lowest risks of injuries, other than in the case of players with very low (<2 SD) acute:chronic ratios (table 2).[13] In a subsequent rugby league study, the odds of sustaining an injury during the competitive season were decreased by 17% for every 10 preseason sessions that players completed, controlling for in-season workloads.[67]

As extensively discussed in Gabbett's recent review,[61] these findings support the notion that workloads should be high enough to produce beneficial physiological adaptations that may

Table 2 Relationships between injury risk and various acute:chronic workloads combined with high and low chronic workloads

Acute:chronic workload ratio	Combined with low chronic workload	Combined with high chronic workload
	Risk of injury (%) ± 90% CI	
Very-low	0.0±0.0	0.0±0.0
Low	7.8±4.1	9.6±4.1
Moderate-low	10.0±2.5	7.5±2.2
Moderate	9.3±2.6	6.2±2.2
Moderate-high	11.0±4.9	7.1±4.0
High	5.9±7.3	12.0±10.7
Very-high	18.2±14.9	28.6±18.1

Risk of injury (%) ± 90% CI associated with 2-week average acute:chronic workload ratios combined with low (<16 095 m) chronic workload, and high (>16 095 m) chronic workload in elite rugby league players. With permission, from Hulin *et al*.[13]

protect against injuries. However, these beneficial effects can be negated if athletes are exposed to large workload spikes, which are likely responsible for a large proportion of non-contact soft tissue injuries.[61] Therefore, the collective findings of workload-injury data indicate that while increased loads are somewhat associated with increased injury rates, optimal load management to minimise injuries entails the accumulation of high chronic workloads, while minimising spikes by maintaining week-to-week changes within ~10%.[61] What remains to be explicated are the mechanisms by which these findings link to our current knowledge of injury aetiology.

AN UPDATED INJURY AETIOLOGY MODEL—INCORPORATING THE EFFECTS OF WORKLOADS

Previous injury aetiology models neither included workloads within the model, nor explained the strong association of loads with injuries. It should be clear that within these models the application of training or competition loads are not inciting events that directly cause injury. An athlete may have a large spike in training and subsequently rupture his or her anterior cruciate ligament, but this injury still occurs via some inciting event (eg, a dynamic valgus collapse). Overuse injuries, or 'training load error injuries',[68] may be strongly workload-related, but still have an inciting event of cumulative tissue overload, even though a specific event occurrence may not be identifiable.[69 70]

Further, workloads are neither a characteristic of the athlete (internal risk factor), nor an aspect of the environment in which the athlete participates (external risk factor). Rather, training and competition loads are better understood as the 'vehicle' in which athletes are exposed to external risk factors and potential inciting events. With this understanding, injuries are not directly caused by workloads. Instead, training and competition loads contribute to injury risk through exposing athletes to potentially injurious situations, as well as through their positive and negative effects on numerous modifiable internal risk factors. Therefore, we propose an updated injury aetiology model that explicitly incorporates workloads within the causal chain, and outlines its known effects (see figure 6).

Like Meeuwisse *et al*'s[10] dynamic recursive model, our model depicts the ever-changing nature of an athlete's injury risk, as well as the possibility that an athlete may, or may not, experience an injury after being exposed to training or competition loads. We have retained the role of internal and external risk factors in line with previous model iterations.[9 24] However, we

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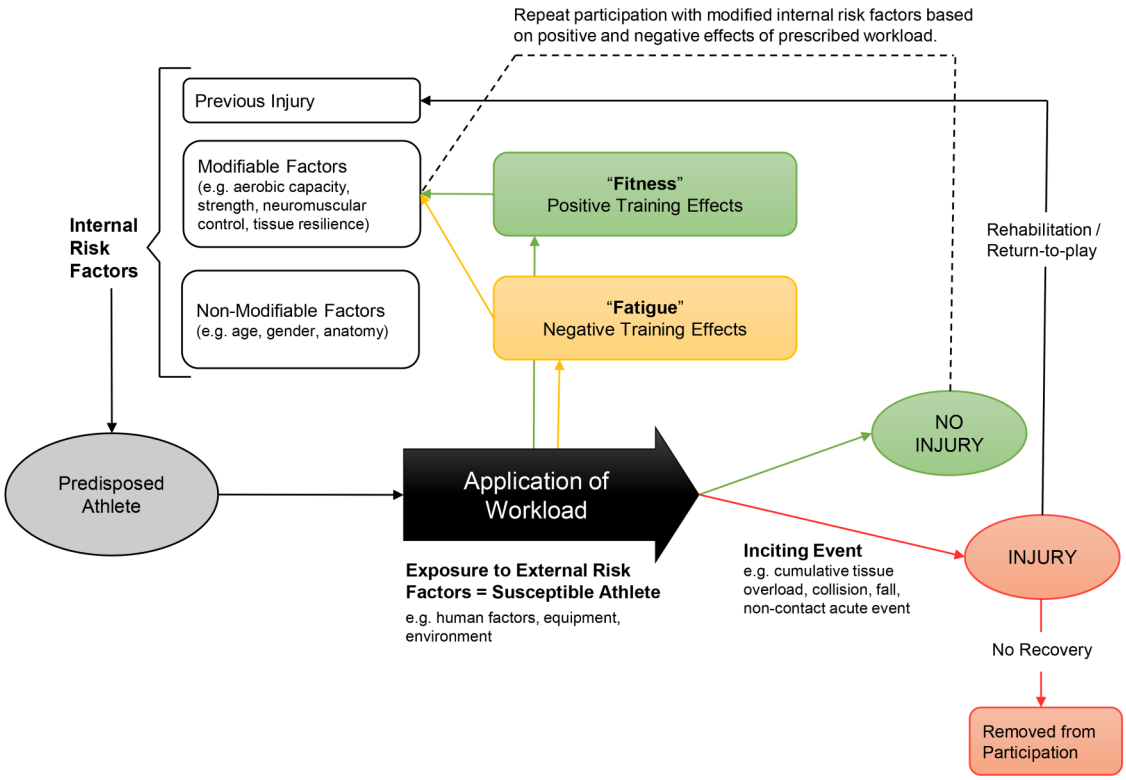

Figure 6 The workload—injury aetiology model.

have divided internal risk factors into modifiable and non-modifiable factors to differentiate those which can change through adaptations from those that are stable. Further, we have extended previous models by including the 'Application of Workload' as the primary process whereby an athlete is exposed to various external risk factors and potential inciting events—the vehicle whereby an athlete moves from being a 'predisposed athlete' to a 'susceptible athlete'.

Whereas Meeuwisse *et al*'s[10] model specified that an athlete who engages in repeat participation may experience adaptation/maladaptation from training and thereby be at an altered risk, we explicitly outline that athlete adaptation comes as the result of each applied workload. As in Banister's original performance prediction models, we highlight that are positive adaptations from training (ie, fitness), and temporary negative effects of training (ie, fatigue). Whereas Banister *et al* focused on these adaptations from a performance perspective, we contend they function similarly in dynamically changing injury risk. These adaptations effect athletes' modifiable internal risk factors so that when they engage in a subsequent training or competition bout, their injury predisposition will be altered.

Lastly, we note that when athletes experience an injury, they must engage in the rehabilitation and return-to-play (RTP) process. In this scenario, their subsequent participation occurs with a modified injury predisposition due to the presence of an index injury.[71] However, in rehabilitating and preparing to return-to-play, these athletes rely on the application of workloads through the same process to restore resilience in the injured tissue and prepare them for the demands of training and competition.[64]

The triple role of workloads—'exposure', 'fitness', and 'fatigue'

Within the updated model, predisposed athletes become susceptible to injury when they are exposed to a training or competition load. This workload exposes them to external risk factors and potential inciting events for the duration of the bout. Further, workloads modify subsequent injury risk. This risk modification occurs through positive and negative adaptations, dictated by the total and relative workload applied. Therefore, workloads affect injury aetiology in three ways.

1. Exposure—training and competition loads are the manner in which athletes are exposed to external risk factors and through which they are exposed to potential inciting events.
2. Fitness—positive adaptations associated with training, which improve modifiable internal risk factors, such as aerobic capacity, skill level, or body composition.
3. Fatigue—negative consequences associated with training, temporarily causing decreased capacity in modifiable internal risk factors, such as tissue resilience or neuromuscular control.

How do workloads influence injury risk? Implications of the updated model

According to this model, workloads are most likely associated with injuries under three main conditions:

1. *High workloads increase exposure, thereby increasing injury risk.*

Since training and competition loads are the means by which athletes are exposed to potential inciting events and external risk factors, there will always be, to some degree, a higher load—higher injury relationship. No load, and thus no exposure, cannot result in athletic injury, regardless of the internal and external risk factors at play. This aspect of the model explains a great deal of the literature which has found increased injury risks with increased workloads.[19 20 50–52]

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2. *Workloads which induce high levels of negative changes to modifiable internal risk factors (ie, 'fatigue') increase injury risk.*

A given external workload is a poor predictor of fatigue, since individuals vary widely in their internal response to set loads. A better marker of how workload effects fatigue may be the acute:chronic workload ratio, wherein a high ratio indicates that athletes have been placed under a load that is substantially higher than their recent training has prepared them for. Physiological and psychological measures of internal load can also indicate athletes' fatigue levels. In this case, negative changes may occur to a number of modifiable internal risk factors, such as compromised neuromuscular control or reduced tissue resilience, which increase subsequent injury predisposition.

3. *Workloads that maximise positive adaptations while minimising fatiguing effects will help make athletes more robust to injury.*

On the other hand, positive adaptations to modifiable internal risk factors help to explain the training load—injury prevention paradox. As long as week-to-week changes and the acute:chronic workload ratio are kept moderate, implicating lower fatiguing effects, higher chronic workloads may protect against injury due to the positive adaptations of training.[61] From a biomechanical perspective, Bahr and Krosshaug[9] explain 'improved fitness may protect the tissue against injury through the effects of training on its material properties'. Together, acknowledging the recursive nature of workload adaptations and its associated positive and negative adaptations helps to frame why the acute:chronic workload ratio is strongly associated with injury, and why high chronic workloads may protect against injury.[12 13 61]

Model limitations

As with any model, our proposed injury aetiology model is incomplete. A set workload may induce various levels of positive and negative adaptations that are based on additional athlete stressors (eg, travel, lack of sleep), which are not explicitly incorporated into the model. Some known risk factors are not explicitly listed (eg, genotype, psychological state, muscle fibre type, etc) and some risk factors are yet to be identified. As just one example, workloads may effect certain psychological variables (eg, commitment to training, self-blame, perceived stress/fatigue, etc), which could subsequently alter match/training behaviour and change injury risk.[48 72] Moreover, the interactions/confounding relationships between various risk factors are not detailed. Some of these, such as the interaction between previous injury and neuromuscular control are known,[9] while others are yet to be explained.

Workloads are unlikely to contribute equally to all injuries, on the basis of different tissue characteristics and mechanisms of injury.[73] Different tissues may respond differently to loads, exemplified in cricket fast bowlers, where high medium term (3-month) workloads are protective against tendon injuries, but a risk factor for bone stress injuries.[73]

Further, workloads may contribute less to contact injuries than non-contact injuries. For example, workloads relate to an ice hockey player's concussion following an illegal body check primarily by providing the exposure to certain external risk factors (eg, opponent behaviour, a compromised helmet). Conversely, a non-contact injury such as a soccer player's ACL rupture while changing direction is likely attributable to a greater number of internal risk factors. In this instance, workloads contribute again in this scenario through exposure, but loads in the weeks preceding injury may also have modified certain internal risk factors—tissue resilience, neuromuscular control, etc—which further predisposed the athlete to injury. Therefore, a number of investigations relating load and injury have focused on non-contact soft-tissue injuries, which are considered 'preventable', more sensitive to load changes, and more attributable to internal risk factors.[11 61]

Similarly, workload may contribute differently to overuse versus acute injuries. Whereas workloads may modify athletes' internal risk such that they are predisposed to an inciting injury event, overuse injuries may occur without a single identifiable injury event. However, the suggested model indicates that this type of cumulative tissue overload could occur with exceedingly high workloads, or spikes in workloads, that result in high levels of fatigue (negative adaptations) without adequate recovery for positive adaptations.

WHERE TO FROM HERE—RESEARCH IMPLICATIONS

The dynamic nature of the current model presents similar challenges to research design and statistical analysis as Meeuwisse *et al*'s[10] dynamic recursive model. From a design perspective, although non-modifiable risk factors can be measured once (since they will not change), modifiable risk factors should be measured repeatedly to account for changes over time. Repeated measurements are even more important in the case of training and competition loads, since quantifying total loads, week-to-week changes, and acute:chronic workload ratios can only be done if workloads are consistently monitored over time. Although repeated measurement presents logistical and financial challenges, repetition is necessary to quantify the dynamic nature of modifiable factors, workloads, their interactions and athletes' subsequent injury risk.

From a statistical analysis perspective, repeated measurement also implicates certain statistical analyses. Simple logistic regression models which assume the same exposure (training and competition load) across individuals may not be appropriate,[24] given the large variability in workload measures across individual players. Although Cox proportional hazards regression (time-to-injury event) and Poisson regression (rate of injury per 1000 exposures or per 1000 hours) have been presented as potential models to account for time exposure,[10] they may fail to account for the intensity of training which is captured in various load measurements. Though not frequently used, multilevel (ie, mixed) modelling approaches may allow for these analyses, since they can account for correlated outcomes (repeated measures among players), and include random effects to predict individual athletes' risks.[74–76] The frailty model which also allows random effects for players and the ability to control for the dependencies of recurrent injuries and repeated measures also shows promise.[77 78]

SUMMARY

Finally, advanced modeling techniques such as Bayesian dynamic probability models, stochastic time-series modeling, and others have also been proposed.[80] Previous injury aetiology models failed to incorporate training and competition workloads. Workloads are neither an internal nor external risk factor for injury, but are the 'vehicle' by which athletes either compete in, or train for their respective sports. We propose that our updated injury aetiology model helps provide a conceptual framework for 'why' workloads are strongly associated with injuries. According to this model, increased workloads increase injury risk through exposure to external risk factors and potential inciting events.[81] However, they also continually modify

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injury predisposition by inducing physiological adaptations, some positive and others negative. Therefore, when athletes experience a spike in training they are not prepared for (ie, a high acute:chronic workload ratio) they likely experience larger degrees of negative maladaptation, modifying a host of internal risk factors and increasing their predisposition to injury in subsequent bouts. Conversely, the accumulation of high chronic workloads while avoiding training spikes maximise the positive physiological adaptations to training and thereby reduce injury risk, as suggested by the training load–injury prevention paradox model.[61]

We note that the confounding and interacting effects of workloads and many risk factors are still unknown. However, longitudinal investigations with repeated measures of workloads and modifiable risk factors may begin to unravel these relationships and provide further insight into the dynamic nature of injury aetiology.

What are the key findings?

- Existing injury aetiology models have not accounted for the effects of training and competition loads on the dynamic nature of injury risk.
- Total workloads, and more notably spikes in workloads (ie, high acute:chronic workload ratios), are strongly associated with athletic injuries, although few mechanisms have been proposed by which workloads contribute to injury causation.
- Within our updated model, workloads contribute to the dynamic injury risk of an athlete through three mechanisms:
  (1) they constitute an exposure to potential inciting events as well as external risk factors, so that increased loads increase the potential for an athlete to experience an injury,
  (2) they induce 'fatigue', representing negative physiological effects which alter internal risk factors and increase injury risk.
  (3) they induce 'fitness', or positive physiological adaptations which alter internal risk factors positively and thereby reduce subsequent injury risk.

How might it impact clinical practice in the future?

Better understanding how training loads relate to injury may assist clinicians in educating, and treating their patients with athletic injuries.

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Acknowledgements The authors would like to acknowledge and thank Professors Meeuwisse and Bahr, as well as their colleagues, whose work in injury aetiology and previous aetiology models have and continue to shape the injury prevention field. The authors are grateful to the manuscript reviewers for their helpful insights which greatly improved the manuscript. Lastly, the authors would also like to thank Sheralyn Windt, Dr. Ben Sporer, Adriaan and Henriette Windt, the Robinsons and the Giesbrechts for their help and feedback in constructing the visual aspects of the injury aetiology model.

Contributors JW was responsible for the initial formulation of the conceptual/aetiological model. Both authors contributed to writing and critical revision of the manuscript.

Funding JW is a Vanier Scholar funded by the Canadian Institutes of Health Research.

Competing interests Karim Khan, the editor in chief of *BJSM* is the PhD supervisor of Johann Windt and was blinded to all phases of the review process.

Provenance and peer review Not commissioned; externally peer reviewed.

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How do training and competition workloads relate to injury? The workload––injury aetiology model

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Br J Sports Med published online July 14, 2016

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